Volkan Ün



Dresden, Germany

kaggle volkan97

volkn

volkan-un

Scholar

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LANGUAGES

English — Fluent

Turkish — Native/Bilingual

German — Basic

SKILLS

Programming Languages: *Python, JavaScript,* MATLAB, *C++*

ML Frameworks: *PyTorch, TensorFlow, JAX*

Simulation Engines: *PyBullet, IsaacSim, MuJoCo*

Frameworks and Libraries: *NumPy, SB3, scikit-learn, OpenCV, Timm, Transformers, Pandas, Matplotlib, Diffusers,* LangChain

Database: *MongoDB, SQLite*

Backend: *flask, express*

Frontend: *react, HTML, CSS*

DevOps - MLOps: *Docker*

Tools: *Git, Poetry, Conda, UV, LaTex, Weights & Biases (W&B)*

INTERESTS

Go (Baduk), Cycling, Dungeons & Dragons, Photography

WORK EXPERIENCE

Deutsche Telekom Chair of Communication Networks - TU Dresden, Researcher
03/2024 – present | Dresden, Germany

- *Developed a multi-agent reinforcement learning algorithm with a learned communication module for multi-robot motion planning.*
- *Developed an SNN model for continuous control with event-based data.*
- *Developed an SNN model for quadrupedal robot locomotion.*
- *Wrote a book chapter on robot learning.*

Transdenlab, WHK (Research Assistant)
02/2022 – 12/2023 | Dresden, Germany

- *Developed an in-house data handling Python package for UKBioBank.*
- *Developed statistical and machine learning models for understanding various psychological disorders, mainly focused on Anorexia Nervosa.*

SmartAlpha, Software Developer (Machine Learning Developer)
04/2021 – 05/2021 | Ankara, Turkey

- *Developed an image segmentation for ultrasound images for nerve segmentation.*
- *Developed generative image models to be used in our image segmentation model.*
- *Developed a federated learning set-up to be used for both decentralization and privacy in model training.*

PUBLICATIONS

Learned Communication for Continuous Multi-Robot Motion Planning
2027 IEEE/SICE International Symposium on System Integration (SII 2027)

Spiking Actor-Critic for Quadruped Locomotion: Spike Encoding Schemes Compared
The 23rd International SoC Conference (ISOCC 2026) - ***Earned ETRI Best Paper Award***.

A Hybrid Spiking Neural Network for Event-Based Continuous Robot Control
2026 IEEE 39th International System-on-Chip Conference (SOCC 2026)

EDUCATION

Computational Modeling and Simulation (M.Sc), Technische Universität Dresden
10/2021 – 03/2024 | Dresden, Germany
GPA: 1.8 / 5.0 (German grade scale) - Computational Life Sciences

- ***Thesis:*** *Reinforcement Learning Applied to Robotics to Automate Peg Transfer Surgical Training Task*
- ***Research Project****: Surgical Image Editing and Generation using Diffusion Models*
- ***Team Project****: Learning the Language of DNA*
- ***Related Courses****: Computational Cell Biology - Applied Bioinformatics - Particle Methods - Introduction to Spatiotemporal Modeling and Simulation*

Mathematics (B.Sc), Hacettepe University
08/2015 – 02/2021 | Ankara, Turkey
GPA: 2.83 / 4.00 (USA grade scale)
Related Courses*: Functional Analysis - Introduction to Hyperbolic Geometry - PDE's - ODE's - Abstract Algebra - General Topology - Linear Algebra - Differential Geometry*

COMPETITIONS

RSNA Screening Mammography Breast Cancer Detection, Silver Medal Winner (74/1687, top 5%)
11/2022 – 02/2023 | Kaggle, Online
Breast cancer identification in screening mammograms. ConvNextV2, EfficientNetV2, TPU training

Cassava Leaf Disease Classification, Silver Medal Winner (95/3900, top 3%)
11/2020 – 02/2021 | Kaggle, Online
The type of disease classification is present on a Cassava Leaf image. Transfer Learning. OverSampling with Light Augmentations - TPUs with TFRecord files